ME 144L: Two-Can System Modeling and Simulation — Pre-Lab

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1 Theoretical Flow Coefficient Calculations

1.1 Can Identification and Dimensions

Can Set 1 is used throughout this pre-lab. Can 1, which will be set at the top, is the small can and Can 2 is the large can placed below Can 1. The measured dimensions are shown in Table 1.

Table 1: Can Set 1 Measured Dimensions

Can	**Role**	**Inner Dia. D (in)**	**Orifice Dia. d (in)**	**Height H (in)**
Can 1 (small)	Top	3.007	0.123	4.315
Can 2 (large)	Bottom	4.005	0.249	5.355

All calculations for orifice flow and volumetric rates use SI units with the conversion $1\,\text{in} = 0.0254\,\text{m}$.

1.2 Ideal Flow Coefficient Derivation

The lecture appendix derives the volumetric flowrate through an orifice as

$$Q = K_o A_o \sqrt{\frac{2P}{\rho}} \tag{1}$$

where $A_o = \pi d^2/4$ is the orifice area and K_o lumps the discharge and velocity coefficients. For the ideal case we are testing, the equation for K_o becomes:

$$K_{o,\text{ideal}} = \frac{1}{\sqrt{1 - \left(\frac{A_o}{A_{\text{can}}}\right)^2}} \tag{2}$$

and the ideal can coefficient, K_{ideal}, yields the state-equation flow coefficient:

$$K_{\text{ideal}} = K_{o,\text{ideal}} \cdot A_o \cdot \sqrt{\frac{2g}{A_{\text{can}}}} \tag{3}$$

From this K_{ideal} is represented in units of $\text{m}^{3/2}\,\text{s}^{-1}$, so that $Q = K_{\text{ideal}}\sqrt{V}$ is consistent dimensionally.

1.3 Numerical Calculation for Can 1 (Small, Top)

Converting dimensions to SI and computing cross-sectional areas:

$$D_1 = (3.007)(0.0254) = 0.07638\,\mathrm{m}$$
$$d_1 = (0.123)(0.0254) = 0.003124\,\mathrm{m}$$
$$A_{1,\mathrm{can}} = \frac{\pi D_1^2}{4} = \frac{\pi(0.07638)^2}{4} = 4.582 \cdot 10^{-3}\,\mathrm{m}^2$$
$$A_{1,\mathrm{o}} = \frac{\pi d_1^2}{4} = \frac{\pi(0.003124)^2}{4} = 7.666 \cdot 10^{-6}\,\mathrm{m}^2$$

With dimensional geometries, we can determine the ideal Can and Orifice coefficients:

$$K_{o,\mathrm{ideal}} = \frac{1}{\sqrt{1 - \left(\frac{7.666 \cdot 10^{-6}}{4.582 \cdot 10^{-3}}\right)^2}} = 1.0000013 \approx 1$$
$$K_1 = A_{1,o}\sqrt{\frac{2g}{A_{1,\mathrm{can}}}} = (7.666 \cdot 10^{-6})\sqrt{\frac{2(9.81)}{4.582 \cdot 10^{-3}}} = 5.017 \cdot 10^{-4}\,\mathrm{m}^{3/2}\,\mathrm{s}^{-1}$$

1.4 Numerical Calculation for Can 2 (Large, Bottom)

$$D_2 = (4.005)(0.0254) = 0.10173\,\mathrm{m}$$
$$d_2 = (0.249)(0.0254) = 0.006325\,\mathrm{m}$$
$$A_{2,\mathrm{can}} = \frac{\pi D_2^2}{4} = 8.128 \cdot 10^{-3}\,\mathrm{m}^2$$
$$A_{2,o} = \frac{\pi d_2^2}{4} = 3.142 \cdot 10^{-5}\,\mathrm{m}^2$$
$$K_2 = A_{2,o}\sqrt{\frac{2g}{A_{2,\mathrm{can}}}} = (3.142 \cdot 10^{-5})\sqrt{\frac{2(9.81)}{8.128 \cdot 10^{-3}}} = 1.544 \cdot 10^{-3}\,\mathrm{m}^{3/2}\,\mathrm{s}^{-1}$$

1.5 Summary of Ideal Flow Coefficients

Table 2: Ideal Flow Coefficients for Can Set 1

Can	A_{can} (m^2)	A_o (m^2)	$K_{o,\mathrm{ideal}}$	K_{ideal} (m$^{3/2}$/s)
Can 1 (top)	$4.582 \cdot 10^{-3}$	$7.666 \cdot 10^{-6}$	1.000001	$5.017 \cdot 10^{-4}$
Can 2 (bottom)	$8.128 \cdot 10^{-3}$	$3.142 \cdot 10^{-5}$	1.000007	$1.544 \cdot 10^{-3}$

Because A_o is significantly less than A_{can} for both cans, the area ratio correction $K_{o,\mathrm{ideal}} \approx 1$. $K_2 > K_1$ because the larger orifice of Can 2 drives a proportionally greater flow per unit, meaning Can 2 drains approximately 3× faster than Can 1.

2 Two-Can Simulation

2.1 System State Equations

The two-can state equations are derived by applying mass continuity to each can independently. With Can 1 as the top (no inflow) and Can 2 as the bottom (fed by Can 1's outflow):

$$\dot{V}_1 = -K_1\sqrt{V_1} \tag{4}$$
$$\dot{V}_2 = +K_1\sqrt{V_1} - K_2\sqrt{V_2} \tag{5}$$

These are nonlinear ODEs and cannot be integrated analytically for the two-can case; numerical integration is required. The simulation uses `scipy.integrate.odeint` over $t \in [0, 120]\,\mathrm{s}$ with 5,000 time steps. Initial conditions are Can 1 filled to capacity ($V_{1,0} = A_{1,\mathrm{can}} \cdot H_1 = 502.2\,\mathrm{mL}$) and Can 2 empty ($V_{2,0} = 0$).

2.2 Simulation Code

The Python simulation is listed below. The ideal K values from Section 1 are set directly and will be replaced with experimentally determined values following the lab.

```python
# Two-can system simulation
# ME 144L Lab - updated spring 2026

import math
import numpy as np
import matplotlib.pyplot as plt

def main():

    # Set initial volumes for can 1 and can 2
    V1o, V2o = 502, 0  # units of milliliters

    # Set simulation time parameters
    dt, t0, tf = 0.001, 0.0, 120
    N = math.floor(abs(tf-t0)/dt)
    x0 = np.array([V1o, V2o])
    t = np.linspace(0, tf, N)

    # Set K1 and K2 with proper units
    # Since Q = K*sqrt(V), then if Q is in ml/s
    # then K must have units of sqrt(ml)/s
    # Placeholder experimental values (replace after lab):
    K1, K2 = 0.501655, 1.543574

    sol1 = rk4fixed(two_can, x0, t, args=(K1, K2))
    V11 = sol1[:,0]
    V21 = sol1[:,1]

    # compute outputs
    Q11, Q21 = [], []
    for i in range(len(t)):
        _, y = two_can(sol1[i,:], t[i], K1, K2)
        Q11.append(y[0])
        Q21.append(y[1])

    print("Peak volume in V2 is :" + str(max(V21)))

    # run with ideal values
    # Can Set 1: K1 (small can, top), K2 (large can, bottom)
    # Ideal K values computed from geometry (SI), converted to sqrt(mL)/s
    K1, K2 = 0.501655, 1.543574

    sol2 = rk4fixed(two_can, x0, t, args=(K1, K2))
    V12 = sol2[:,0]
    V22 = sol2[:,1]

    # compute outputs
    Q12, Q22 = [], []
    for i in range(len(t)):
        _, y = two_can(sol2[i,:], t[i], K1, K2)
        Q12.append(y[0])
        Q22.append(y[1])

    print("Peak volume in V2-ideal is :" + str(max(V22)))

    fig, (ax1, ax2) = plt.subplots(2)
    fig.suptitle('Two-can simulation')
    ax1.plot(t, V11, c='b', label='V1')
    ax1.plot(t, V21, c='b', label='V2')
    ax1.plot(t, V12, c='r', linestyle='dashed', label='V1-ideal')
```

```python
    ax1.plot(t, V22, c='r', linestyle='dashed', label='V2-ideal')
    ax1.legend(loc="upper right")
    ax2.plot(t, Q11, c='b', label='Q1')
    ax2.plot(t, Q21, c='b', label='Q2')
    ax2.plot(t, Q12, c='r', linestyle='dashed', label='Q1-ideal')
    ax2.plot(t, Q22, c='r', linestyle='dashed', label='Q2-ideal')
    ax2.legend(loc="upper right")
    plt.xlabel('Time (sec)')
    plt.grid()
    plt.show()


def two_can(x, t, K1, K2):
    V1, V2 = x[0], x[1]
    if V1 <= 0.0:
        Q1 = 0
    else:
        Q1 = K1*np.sqrt(abs(V1))
    if V2 <= 0.0:
        Q2 = 0
    else:
        Q2 = K2*np.sqrt(abs(V2))
    y = [Q1, Q2]
    return np.array([-Q1, +Q1-Q2]), y


def rk4fixed(f, x0, t, args=()):
    import numpy
    n = len(t)
    x = numpy.zeros((n, len(x0)))
    x[0] = x0
    for i in range(n - 1):
        h = t[i+1] - t[i]
        k1, _ = f(x[i], t[i], *args)
        k2, _ = f(x[i] + k1 * h / 2., t[i] + h / 2., *args)
        k3, _ = f(x[i] + k2 * h / 2., t[i] + h / 2., *args)
        k4, _ = f(x[i] + k3 * h, t[i] + h, *args)
        x[i+1] = x[i] + (h / 6.) * (k1 + 2*k2 + 2*k3 + k4)
    return x
main()
```

Listing 1: two_can_sim.py — Two-Can Hydraulic System Simulation

2.3 Simulation Results

Figure 1 shows the simulated height and volume histories for Can 1 initially full and Can 2 initially empty, using ideal K values.

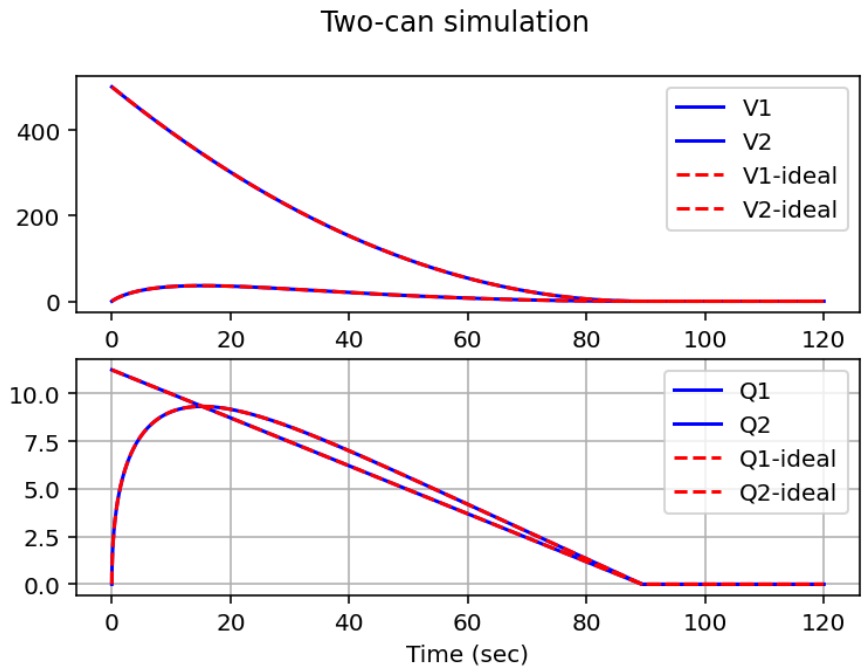

Figure 1: Two-can simulation using ideal K values (Can Set 1). Top panel: fluid height in each can. Bottom panel: volume.

The simulation tracks volume V and flow rate Q for each can over 120 s using an RK4 integrator with $dt = 0.001\,\mathrm{s}$. Initial conditions are Can 1 filled to capacity ($V_{1,0} = 502\,\mathrm{mL}$) and Can 2 empty ($V_{2,0} = 0$). Experimental K values will replace the ideal placeholders following the lab.

3 Lab Procedure for Determining Coefficients Experimentally

The K values for each can are determined from single-can emptying experiments. Integrating the single-can ODE $\dot{V} = -K\sqrt{V}$ from initial volume V_o to empty over time T_e yields:

$$K = \frac{2\sqrt{V_o}}{T_e} \tag{6}$$

Running multiple trials at different initial volumes and fitting $\sqrt{V_o}$ vs. $T_e/2$ through the origin via linear regression gives a robust estimate of K. This procedure is performed on each can independently.

Equipment required: cans removed from two-can rig, graduated beakers (multiple sizes), collection beaker, stopwatch or phone timer.

1. Remove the target can from the two-can test set up. Then place it above a collection beaker with orifice pointing downward.
2. Choose 4 to 5 initial volumes spanning a representative range (e.g., 50, 100, 200, 300, 400 mL for Can 1; scale proportionally for Can 2). Measure with a graduated beaker.
3. Pour the target volume V_o into the can while covering the orifice with a fingertip, the write down the value of V_o in mL.
4. Simultaneously remove the finger from the orifice and start a stopwatch.
5. Stop when the last visible stream of water exits the orifice. Record the emptying time T_e in seconds.
6. Run a minimum of 2 trials per initial volume and average T_e to reduce timing uncertainty.
7. Convert all V_o readings from mL to m^3 ($1\,\mathrm{mL} = 1 \cdot 10^{-6}\,\mathrm{m}^3$).

8. Apply Equation 6 to each averaged (V_o, T_e) group.
9. Plot $\sqrt{V_o}$ vs. $T_e/2$ and fit a line forced through the origin. The slope equals K. Record the slope and R^2 value.
10. Perform Steps 1–9 on the other can to determine its K independently.
11. Replace the experimental `K1` and `K2` placeholders in `two_can_sim.py` with the measured values (in units of $\sqrt{\mathrm{mL}}\,\mathrm{s}^{-1}$) and re-run before the two-can validation test.